

Particle=excitation of field

Field

Equation of motion

$$(\partial_t^2 - c^2 \nabla^2 + m^2)\phi = 0$$

Field kick

$$\phi(0, r) = A e^{-r^2/(2\sigma^2)}, \quad \dot{\phi}(0, r) = 0$$

$$\phi(t, r) = A\sigma^2 \int_0^\infty k dk J_0(kr) e^{-\sigma^2 k^2/2} \cos\left(t \sqrt{c^2 k^2 + m^2}\right)$$

Field wave evolution

